THE DUAL MODULE IS PURE INJECTIVE AND EVERY MODULE EMBEDS UNIVERSALLY INTO A PURE INJECTIVE

THE STACKS PROJECT AUTHORS

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The properties of the duality functor M -> Hom_Z(M, Q/Z). More on Algebra, Section more-algebra-section-injectives-modules, invoked at proof lines 2729-2732. Supplies the adjunction, exactness and faithfulness used throughout the proof.
- Q/Z is injective as an abelian group. Standard property of divisible abelian groups, used at proof lines 2735-2736. Makes M^vee pure injective.

Source reference. Original section 01D8: sources/stacks-chow/tags.txt, source lines 1789–1789. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. For any A-module M set $M^\vee = \mathrm{Hom}_{\mathbf{Z}}(M, \mathbf{Q}/\mathbf{Z})$.*

1. *For any A-module M the A-module $M^\vee$ is pure injective.*
2. *An A-module I is pure injective if and only if the map $I \to (I^\vee)^\vee$ splits.*
3. *For any module M there exists a universally exact sequence $0 \to M \to I \to N \to 0$ with I pure injective.*

Proof. We will use the properties of the functor $M \mapsto M^\vee$ found in More on Algebra, Section 01D8 without further mention. Part (1) holds because $\mathrm{Hom}_A(N, M^\vee) = \mathrm{Hom}_{\mathbf{Z}}(N \otimes_A M, \mathbf{Q}/\mathbf{Z})$ and because $\mathbf{Q}/\mathbf{Z}$ is injective in the category of abelian groups. Hence if $I \to (I^\vee)^\vee$ is split, then I is pure injective. We claim that for any A-module M the evaluation map $ev : M \to (M^\vee)^\vee$ is universally injective. To see this note that $ev^\vee : ((M^\vee)^\vee)^\vee \to M^\vee$ has a right inverse, namely $ev' : M^\vee \to ((M^\vee)^\vee)^\vee$. Then for any A-module N applying the exact faithful functor $^\vee$ to the map $N \otimes_A M \to N \otimes_A (M^\vee)^\vee$ gives

$$\mathrm{Hom}_A(N, ((M^\vee)^\vee)^\vee) = \Big(N \otimes_A (M^\vee)^\vee\Big)^\vee \to \Big(N \otimes_A M\Big)^\vee = \mathrm{Hom}_A(N, M^\vee)$$

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which is surjective by the existence of the right inverse. The claim follows. The claim implies (3) and the necessity of the condition in (2). □